

HAFIY HARIZAN

Software Engineer

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SUMMARY

Software engineer with 5+ years designing, building and running production data platforms, ETL pipelines and backend services. Works primarily in Python, PHP and SQL, with JavaScript and TypeScript front ends, and ships through Docker and GitLab CI/CD.

At Telekom Malaysia, built platforms that processed operational network data for 3M+ customers and automated regulatory compliance reporting end to end. At WA Health, engineers data feeds that integrate multiple HR systems and led a 12-analyst team's move to Git-based version control. Focused on clean, maintainable code and reliable automation, and looking for a mid to senior software engineering role on a data platform team in Perth.

TECHNICAL SKILLS

Languages: Python, SQL, PHP, JavaScript, TypeScript, Bash

Data: Pandas, NumPy, FME, Power BI (DAX), Tableau

Databases: Oracle, PostgreSQL, MySQL, MariaDB

Backend & web: FastAPI, Node.js, React, Next.js, Leaflet.js, GeoServer

DevOps & cloud: Docker, Git, GitHub, GitLab CI/CD, Linux, Azure (OpenAI, Machine Learning)

PROFESSIONAL EXPERIENCE

Business Systems Administrator (Data Analyst) — Health Support Services, WA Health **June 2026 – Present**

- Design and build a daily data feed that consolidates approved extended leave (30+ days) from three WA Health HR systems, so ICT can pause Microsoft 365 licences and park Teams numbers automatically. Wrote the business requirements and the leave-period logic, including working-day gap rules.
- Designed the GitHub organisation for a 12-analyst team: an eight-repository structure, topic tagging, and a version control strategy for Power BI using the text-based .pbip format. Took it from a shared-drive pilot to adoption, with an interactive onboarding guide and internal docs on GitHub Pages.
- Write SQL against Oracle HR data warehouses and build data models for workforce analytics, covering headcount, FTE, turnover, leave and recruitment.
- Automated data preparation, validation and refresh steps with Python, reducing manual handling and improving the reliability of recurring reporting.
- Build complex Power BI models and DAX for senior leadership and operational teams, and support the team's reporting platforms and tools.

Visualisation & Software Engineer — Telekom Malaysia **March 2024 – June 2025**

- Built data platforms in PHP, JavaScript and SQL that processed operational data from 3M+ customers, improving system stability and speeding up enterprise reporting.
- Automated data ingestion and validation workflows and integrated machine learning outputs into operational dashboards for network diagnostics.
- Applied version control, modular architecture and documentation standards to keep production systems maintainable and secure.
- Added validation logic, access controls and monitoring scripts to support high availability and regulatory compliance.
- Built real-time infrastructure web apps and dashboards with AmCharts and GeoServer, replacing manual reporting.

Network Geospatial Visualisation Solution Engineer — Telekom Malaysia **March 2022 – March 2024**

- Automated spatial data processing pipelines with FME and Bash, cutting manual effort for engineering and analytics teams.
 - Deployed and managed GeoServer services and Tableau Server, providing secure, reliable access for internal users.
 - Tuned PostgreSQL and MariaDB performance through indexing, query optimisation and schema restructuring for large mapping workloads.
 - Built geospatial tools and dashboards with Power BI, Tableau and GeoServer for network planning.
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KEY PROJECTS

- **HRPlus Operations Data Pipeline & Dashboard (WA Health)** — Wrote SQL to extract HR service transactions from Oracle and transformed them into a Power BI data model. Encoded the business rules for backlog, queue ageing and service levels as reusable DAX measures, giving operations leaders one source of truth for form volumes, processing trends and SLA performance.
 - **NurseWest Business Continuity Automation (WA Health)** — Python automation that extracts, transforms and delivers the data behind NurseWest's business continuity processes. Replaced manual data handling so operational and workforce outputs are produced reliably and on time.
 - **DREAM (Data Repository for Exploratory Analysis and Management)** — A central data platform with automated ingestion and preprocessing pipelines behind a Python FastAPI service layer and a Next.js front end. Cut data retrieval from hours to minutes, serving analysts and engineers clean, analysis-ready datasets on demand.
 - **NDM (Network Data Mart)** — A batch ETL data mart with cron-scheduled pipelines (PHP, JavaScript, MariaDB) ingesting and transforming millions of records a day into reporting tables. Fully automated reports that operational teams previously built by hand.
 - **MSQoS (Mandatory Standards for Quality of Service)** — A regulatory data pipeline for MCMC reporting, with validation checks, monitoring scripts and secure data delivery. Achieved 100% on-time submissions and cut manual intervention by about 50%.
 - **FIVE (Fibre Infrastructure Visualisation and Enhancement)** — A spatial data pipeline that ingests and stores GeoPackage datasets through FME and Bash, feeding ML-based shortest-path routing on a Leaflet web map. Improved route-planning accuracy by about 20%.
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EDUCATION & CERTIFICATION

- **Master of Data Science** — Universiti Teknologi MARA (UiTM), Malaysia, 2021–2022
- **Bachelor of Mechatronics Engineering (Honours)** — Universiti Teknikal Malaysia Melaka (UTeM), Malaysia, 2015–2019
- **Microsoft Certified: Azure AI Engineer Associate** — issued March 2026, expires March 2027